Table S7 Random-seed and receptor-structure sensitivity

A. Five-seed distributions

Target pair	Metric	Seed 42	Median	Minimum	Maximum	IQR	Finite seeds
EGFR/HER2	AUROC B	0.6274	0.6375	0.6274	0.6477	0.0111	5
EGFR/HER2	AUROC A	0.2467	0.2388	0.2305	0.2467	0.0089	5
EGFR/HER2	Minimum AUROC	0.2467	0.2388	0.2305	0.2467	0.0089	5
JAK1/JAK2	AUROC B	0.5908	0.5840	0.5684	0.5908	0.0093	5
JAK1/JAK2	AUROC A	0.7485	0.7485	0.7422	0.7573	0.0112	5
JAK1/JAK2	Minimum AUROC	0.5908	0.5840	0.5684	0.5908	0.0093	5
JAK1/TYK2	AUROC B	0.6475	0.6475	0.6328	0.6631	0.0156	5
JAK1/TYK2	AUROC A	0.4219	0.4229	0.4121	0.4302	0.0015	5
JAK1/TYK2	Minimum AUROC	0.4219	0.4229	0.4121	0.4302	0.0015	5
PIK3CA/mTOR	AUROC B	0.7363	0.7354	0.6997	0.7402	0.0029	5
PIK3CA/mTOR	AUROC A	0.5801	0.5752	0.5596	0.5869	0.0190	5
PIK3CA/mTOR	Minimum AUROC	0.5801	0.5752	0.5596	0.5869	0.0190	5
AChE/BChE	AUROC B	0.6341	0.6422	0.6049	0.6533	0.0163	5
AChE/BChE	AUROC A	0.6442	0.6442	0.6356	0.6448	0.0026	5
AChE/BChE	Minimum AUROC	0.6341	0.6356	0.6049	0.6448	0.0075	5
F2/F10	AUROC B	0.4082	0.4062	0.3955	0.4082	0.0015	5
F2/F10	AUROC A	0.3623	0.3647	0.3594	0.3682	0.0029	5
F2/F10	Minimum AUROC	0.3623	0.3647	0.3594	0.3682	0.0029	5
PPARG/PPARA	AUROC B	0.6553	0.6553	0.6465	0.6680	0.0127	5
PPARG/PPARA	AUROC A	0.6021	0.6313	0.6021	0.6406	0.0225	5
PPARG/PPARA	Minimum AUROC	0.6021	0.6313	0.6021	0.6406	0.0225	5
PPARA/PPARD	AUROC B	0.6787	0.6929	0.6787	0.6953	0.0039	5
PPARA/PPARD	AUROC A	0.5107	0.5088	0.4800	0.5151	0.0146	5
PPARA/PPARD	Minimum AUROC	0.5107	0.5088	0.4800	0.5151	0.0146	5

Five Vina seeds: 17, 29, 42, 71 and 101. Each seed uses its own valid population. Receptor comparisons use paired common members and replace only one target. Full direction-specific estimates, confidence intervals and members are in the source data. PIK3CA/mTOR Q110 has no alternative-receptor calculation. AUROC, target B compares dual with A-only; AUROC, target A compares dual with B-only. Minimum AUROC is the smaller directional AUROC.

Table S7 Random-seed and receptor-structure sensitivity

B. One-receptor substitutions

Target pair	Alt. PDB	Side	n (dual / A-only / B-only)	Original minimum AUROC	Alternative minimum AUROC	Minimum AUROC difference (95% CI)	Affected AUROC difference	Unreplaced AUROC difference
AChE/BChE	1P0M	B	26 / 23 / 26	0.6037	0.6003	-0.0033 [-0.0803, 0.0836]	-0.0033	0
AChE/BChE	4EY6	A	27 / 23 / 26	0.6184	0.5670	-0.0514 [-0.1852, 0]	-0.0940	0
F2/F10	2Y5F	B	30 / 32 / 31	0.3828	0.3828	0 [-0.0469, 0.0229]	-0.0146	0
F2/F10	3SHC	A	30 / 32 / 31	0.3828	0.4333	0.0505 [0, 0.1441]	0.0780	0
PPARA/PPARD	5U46	B	32 / 32 / 32	0.5107	0.5107	0 [-0.0225, 0]	-0.0181	0
PPARA/PPARD	6KAX	A	32 / 32 / 32	0.5107	0.4438	-0.0669 [-0.1484, 0.0044]	-0.0669	0

Five Vina seeds: 17, 29, 42, 71 and 101. Each seed uses its own valid population. Receptor comparisons use paired common members and replace only one target. Full direction-specific estimates, confidence intervals and members are in the source data. PIK3CA/mTOR Q110 has no alternative-receptor calculation. AUROC, target B compares dual with A-only; AUROC, target A compares dual with B-only. Minimum AUROC is the smaller directional AUROC.